

# Unbounded normalized cofactor spectra for rank two Gram matrices

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## Abstract

For a positive-semidefinite Hermitian matrix  $A$ , let  $C(A)_{ij} = a_{ij}$  per  $A(i | j)$ , where the minor deletes row  $i$  and column  $j$ . We prove that  $\lambda_{\max}(C(A))/\text{per } A$  has no dimension-independent upper bound, already for complex rank-two correlation matrices. The same construction makes  $\lambda_{\max}(\text{Re } C(A))/\text{per } A$  arbitrarily large. The proof computes a spectral limit under repeated Gram vectors and combines it with projective equidistribution. It also gives positive-definite examples and unbounded local second-order relative coefficients for entrywise-positive real correlation perturbations.

## 1 Statement and scope

Our convention is that  $A(i | j)$  deletes row  $i$  and column  $j$ , without transposing this minor. All Hermitian inner products below are linear in their first argument. The matrix  $C(A)$  is positive semidefinite and  $C(A)\mathbf{1} = (\text{per } A)\mathbf{1}$ . Drury disproved the assertion that this prescribed eigenvalue is always the largest one [1]. Here we consider all constant multiples of the permanent.

**Theorem 1.** *For every  $R > 0$  there are an integer  $N$  and a complex Hermitian correlation matrix  $A$  of rank exactly two such that*

$$\lambda_{\max}(C(A)) > R \text{per } A, \quad \lambda_{\max}(\text{Re } C(A)) > R \text{per } A.$$

*There are also positive-definite correlation matrices with both inequalities. The positive-definite assertion does not retain rank two.*

The real part is entrywise, so its largest eigenvalue is the maximum Rayleigh quotient over real unit vectors. The theorem is an unboundedness assertion as the order varies; no rate in  $N$  is asserted. It is distinct from unbounded Hadamard-product bunching ratios [3], and from bounds for the full Schur-power matrix. It does not resolve Lieb's normalized character or subgroup permanental-dominance conjecture [4]. A separate source audit records the finite literature search; no claim of exhaustive priority verification is made.

## 2 A repetition limit

Let  $v_i = (a_i, b_i)$  be nonzero unit rows,  $1 \leq i \leq n$ , put  $\ell_i(x, y) = a_i x + b_i y$ , and set  $P = \prod_i \ell_i$  and  $A_{ij} = v_i v_j^*$ . For binary forms of degree  $d$  define

$$\left\langle \sum_{k=0}^d p_k x^{d-k} y^k, \sum_{k=0}^d q_k x^{d-k} y^k \right\rangle_B = \sum_{k=0}^d (d-k)! k! p_k \overline{q_k}.$$

Leibniz expansion gives

$$\left\langle \prod_i \ell_i, \prod_j \ell'_j \right\rangle_B = \text{per}[\langle \ell_i, \ell'_j \rangle_B]. \quad (1)$$

In particular  $\text{per } A = \|P\|_B^2 > 0$ , and  $C(A)$  is the Gram matrix of  $v_i \otimes (P/\ell_i)$ . This proves its positivity with the stated conventions. For Haar probability measure  $\mu$  on  $\mathbb{CP}^1$ , represented by unit vectors,

$$\langle p, q \rangle_B = (d+1)! \int p \bar{q} d\mu. \quad (2)$$

Indeed, phase integration makes distinct monomials orthogonal and  $\int |x|^{2a} |y|^{2b} d\mu = a!b!/(a+b+1)!$ .

Repeat each row  $L$  times, write  $N = nL$ , and let  $A_L$  be the resulting Gram matrix. Then  $P_L = P^L$  and, with cluster indices  $(i, a)$ ,

$$C(A_L)_{(i,a),(j,b)} = A_{ij} \langle P^L/\ell_i, P^L/\ell_j \rangle_B. \quad (3)$$

Let  $U_L : \mathbb{C}^n \rightarrow \mathbb{C}^{nL}$  map  $e_i$  to the vector equal to  $L^{-1/2}$  on cluster  $i$  and zero elsewhere. All clusters in (3) are constant, so  $C(A_L)$  vanishes on  $\text{ran}(U_L)^\perp$ . Its nonzero normalized spectrum, including multiplicities, is that of  $K_L = U_L^* C(A_L) U_L / \text{per } A_L$ . Writing  $d\mu_L = |P|^{2L} d\mu / \int |P|^{2L} d\mu$ , the factorials in (2) give the exact formula

$$(K_L)_{ij} = \frac{L}{nL+1} A_{ij} \int \frac{d\mu_L}{\ell_i \bar{\ell}_j}. \quad (4)$$

The expression against the density at a zero is interpreted through the polynomial  $(P^L/\ell_i) \overline{(P^L/\ell_j)}$ .

**Lemma 2.** *Suppose  $|P|$  has a unique projective maximum. Rotate it to  $(1, 0)$  and put  $z_i = b_i/a_i$ . Then every  $a_i \neq 0$ ,  $\sum_i z_i = 0$ , and*

$$K_L \longrightarrow \frac{J_n + zz^*}{n} \quad \text{in operator norm.}$$

Consequently

$$\lim_{L \rightarrow \infty} \frac{\lambda_{\max}(C(A_L))}{\text{per } A_L} = \max \left( 1, \frac{\|z\|^2}{n} \right), \quad (5)$$

$$\lim_{L \rightarrow \infty} \frac{\lambda_{\max}(\text{Re } C(A_L))}{\text{per } A_L} \geq \max \left( 1, \frac{\|z\|^2}{2n} \right). \quad (6)$$

*Proof.* The maximum of the nonzero polynomial is positive, so no factor vanishes there. The integrand in (4) is continuous near the maximum  $u$ . Outside any neighborhood of  $u$ ,  $|P| \leq a < |P(u)|$ . On a smaller neighborhood of positive measure,  $|P| \geq b > a$ . The absolute unnormalized tail is bounded by

$$a^{2L-2} \sup |(P/\ell_i)(P/\ell_j)|,$$

whereas the denominator is bounded below by a positive constant times  $b^{2L}$ . Thus the apparent poles cause no problem and the integral tends to  $(a_i \bar{a}_j)^{-1}$ . Since  $n$  is fixed, entrywise convergence implies operator-norm convergence and gives the stated matrix limit.

In the local coordinate  $(1, t)/\sqrt{1+|t|^2}$ ,

$$\log |P|^2 = \text{constant} + \sum_i \log |1 + z_i t|^2 - n \log(1 + |t|^2).$$

The two real first derivatives at zero vanish, hence  $\sum_i z_i = 0$ . The limiting matrix therefore has eigenvalues 1,  $\|z\|^2/n$ , and zeros, which proves (5). For  $z = x + iy$  its real part is  $(J_n + xx^T + yy^T)/n$ , with  $x, y \perp \mathbf{1}$ . The last two terms have rank at most two and trace  $\|z\|^2/n$ , proving (6). Compression commutes with real part because  $U_L$  is real.  $\square$

### 3 Equidistribution and selection of one maximum

Choose collections of  $m$  unit rows whose empirical projective measures converge weakly to Haar measure on  $\mathbb{CP}^1$ . Such a sequence is obtained from midpoint grids in  $p = |a|^2 \in (0, 1)$  and the relative phase, with both grid sizes tending to infinity. Let  $u_m$  be any maximum of the corresponding product, and rotate variables so that  $u_m = (1, 0)$ .

These moving rotations preserve weak convergence to Haar. Indeed, every subsequence of rotations has a convergent subsubsequence by compactness of  $U(2)$ . For each continuous test function, the rotated functions then converge uniformly; weak convergence and Haar invariance give the same limiting integral. This subsequence argument proves the claim for the full sequence.

For  $T > 0$  the function

$$f_T([a : b]) = \min(|b/a|^2, T), \quad f_T([0 : 1]) = T,$$

is continuous. Since  $|a|^2$  is uniform on  $[0, 1]$  under Haar measure,

$$\int f_T d\mu = \int_0^1 \min\left(\frac{1-p}{p}, T\right) dp = \log(1+T). \quad (7)$$

Every  $a_i$  in the rotated configuration is nonzero. It follows that

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \sum_{i=1}^m |b_i/a_i|^2 \geq \log(1+T) \quad \text{for every } T > 0.$$

Hence these averages tend to infinity. Only bounded continuous test functions have been used; there is no assumption of convergence of an unbounded moment.

To make the maximum unique, append one unit linear form  $\ell_u(z) = \langle z, u_m \rangle$ , linear in  $z$ . On the unit sphere its modulus is at most one, with equality only at the projective point  $u_m$ . Multiplication by this factor leaves the old maximum value unchanged and makes its location unique. In the rotated coordinates the added row is  $(1, 0)$ , contributing  $z = 0$ . For the augmented base  $n = m + 1$  we still have

$$\frac{1}{n} \sum_{i=1}^n |z_i|^2 = \frac{1}{m+1} \sum_{i=1}^m |b_i/a_i|^2 \rightarrow \infty. \quad (8)$$

*Proof of Theorem 1.* Given  $R$ , first choose one finite base in (8) with average exceeding  $2R + 2$ . Equidistribution ensures that a sufficiently large base spans  $\mathbb{C}^2$ . Freeze it and then take a sufficiently large integer  $L$  in Lemma 2. Both ratios exceed  $R$ , all rows are unit, and the Gram matrix has rank exactly two.

For a strict witness  $A$ , polynomial dependence of  $C(A)$  and  $\text{per } A$ , continuity of Hermitian eigenvalues, and  $\text{per } A > 0$  show that the strict inequalities persist for  $A + \varepsilon I$  with sufficiently small  $\varepsilon > 0$ . The matrix  $(A + \varepsilon I)/(1 + \varepsilon)$  is a positive-definite correlation matrix. Both ratios are unchanged by this final scalar rescaling, since  $C(cA) = c^N C(A)$  and  $\text{per}(cA) = c^N \text{per } A$ .  $\square$

### 4 A local consequence and an exact finite certificate

The following is the real-direction specialization of the perturbation formula of Pioge and coauthors [2, Lemma 1]; we include the calculation to fix the Gram and conjugation conventions. For a real unit vector  $w$ , let

$$B(\varepsilon)_{ij} = \frac{1 + \varepsilon^2 w_i w_j}{\sqrt{(1 + \varepsilon^2 w_i^2)(1 + \varepsilon^2 w_j^2)}}.$$

This is a real correlation matrix of rank at most two, with strictly positive entries for sufficiently small  $\varepsilon$ . Expanding the entries and then the permanent gives

$$\frac{\text{per}(A \circ B(\varepsilon))}{\text{per } A} = 1 + \varepsilon^2 \left( \frac{w^T \text{Re } C(A)w}{\text{per } A} - 1 \right) + O(\varepsilon^4). \quad (9)$$

In fact the quadratic coefficient before division by  $\text{per } A$  is

$$\sum_{ij} C(A)_{ij} \left( w_i w_j - \frac{w_i^2 + w_j^2}{2} \right) = w^T C(A)w - \text{per } A = w^T \text{Re } C(A)w - \text{per } A,$$

using both row and column sums. Theorem 1 therefore makes the relative coefficient in (9) arbitrarily large. This is a local statement; it gives no uniform finite- $\varepsilon$  gain or perturbative interval. For complex  $w$ , the column-Gram convention in [2] has numerator  $1 + \varepsilon^2 \overline{w_i} w_j$ ; replacing it by the opposite convention would conjugate the Rayleigh direction.

For an optional finite check, the companion file `witness_n200.csv` specifies 200 Gaussian-integer rows  $v_i \in \mathbb{Z}[i]^2$  and a vector  $w \in \mathbb{Z}[i]^{200}$ . The standard-library integer verifier proves

$$\frac{269}{100} < \frac{w^* C(VV^*)w}{\text{per}(VV^*)\|w\|^2} < \frac{27}{10}.$$

It independently forms each omitted-factor polynomial and evaluates  $\sum_{a=1}^2 \|\sum_i \overline{w_i} v_{ia}(P/\ell_i)\|_B^2$ . This certificate illustrates a finite violation and checks the conventions; unboundedness follows from the analytic proof, not from numerical extrapolation.

## References

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- [2] L. Pioge, K. K. Pietrasz, B. Seron, L. Novo and N. J. Cerf, A logical implication between two conjectures on matrix permanents, *Linear Algebra Appl.* **725** (2025), 309–318. doi:10.1016/j.laa.2025.07.011; arXiv:2508.00111.
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